8. A new timer, in the shape of a ball, has been produced. When it is released it starts a built-in stop clock. When the ball next hits something the timer stops. The time taken is displayed on its screen.

A teacher uses the ball timer for an experiment. He holds the ball out of his third floor classroom window and then carefully releases it. The time taken to fall is recorded in a table. The ball is dropped 5 times during the experiment.

Drop number	1	2	3	4	5
Time (s)	1.48	1.10	1.52	1.54	1.46

(a) Explain how the teacher could find out if the results are reproducible. [2]

....

....

....

(b) **Circle** the result in the table which is most likely to be anomalous. [1]

(c) Calculate the mean time for the ball timer to fall. [2]

Mean time to fall = ... s

Examiner only

(d) The mean impact velocity of the ball on the ground is 14.2 m/s.
Use the equation:

$$x = \frac{u + v}{2} t$$

to calculate the drop height of the ball. [2]

Drop height = ... m

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